

A Geometric-Null Binomial Framework for Harmonic Longitude Proximity Testing in Monument Distributions

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Highlights

- Geometric-null binomial is the test matched to harmonic-proximity hypotheses.
- Rayleigh concentration and binomial proximity enrichment can dissociate.
- Dissociation shown on the UNESCO dome/stupa corpus ($N = 90$).
- Worked example: Wiki+OSM Fisher $\chi^2(4) = 28.74$, $p < 0.001$, T3, excluding domes.
- Negative controls (Americas, N. American mounds) show no proximity enrichment.

Abstract

Spatial periodicity analyses of monument distributions routinely apply the Rayleigh statistic to evaluate longitude-grid hypotheses, yet Rayleigh concentration and thresholded harmonic proximity answer different questions and can dissociate within the same corpus. This paper introduces the geometric-null binomial framework as the matched test for repeated-line grid hypotheses: (i) define phases $\theta_i = 2\pi((\lambda_i - \lambda_0) \bmod T)/T$; (ii) compute harmonic deviation $\delta_i = \min(r_i, T - r_i)$; (iii) test the count of sites with $\delta_i \leq d$ against the geometric null $p_0 = 2d/T$; (iv) combine across independently assembled corpora via Fisher's method; and

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(v) apply an anchor sweep correction based on a max statistic when anchor search is evaluated. A period sweep of three openly available hemispherical-monument corpora—UNESCO dome/stupa ($N = 90$), Wikidata Q180987, and OSM stupa layer—illustrates the dissociation empirically and provides a cross-corpus worked example. At $T = 3^\circ$ and the $\sim 35.25^\circ\text{E}$ Levantine corridor phase, the two-corpus combination excluding the period-motivating dome corpus (Wikidata + OSM) gives $\chi^2(4) = 28.74$, $p < 0.001$ at T3 (threshold $\pm 0.10 \times T$); negative controls in the Western Hemisphere show no proximity enrichment. Following [Silva \(2020\)](#), all results are treated as statistical descriptions of the observed coordinate distribution; the analysis does not identify causal or intentional relationships. All data, code, and audit files are publicly available.

Keywords: spatial statistics, circular statistics, harmonic proximity, binomial geometric null, monument distribution, archaeoastronomy

1 Introduction

Do monument coordinates cluster near specific longitude intervals? The question recurs in the archaeoastronomical and archaeogeographic literature, often discussed through observational coincidences involving sacred sites, trade corridors, or cosmographic traditions. This paper focuses on *hemispherical monuments*, defined here as structures whose principal form is a rounded or mounded mass, including Buddhist stupas, corbelled tholos tombs, and domed shrines, as a corpus for testing longitude-grid hypotheses. This broader question has also been criticized on grounds of investigator-assembled site lists, insufficient correction for multiple comparisons, and lack of external comparison ([Williamson & Bellamy, 1983](#); [Freeman, 1976](#); [Kendall, 1974](#)). These concerns motivate a stricter separation between statistical detection and archaeological explanation. Alongside these critiques, this paper raises the complementary issue of matching tests to hypotheses. The standard circular-statistical tool used in this literature measures a different quantity from the one predicted by a repeated-line grid hypothesis.

The tool in question is the Rayleigh statistic R . In its circular-statistical form, Rayleigh’s mean-resultant test is a standard test of departure from uniformity toward a single preferred direction on the circle (Rayleigh, 1919; Fisher, 1993; Mardia & Jupp, 2000), most commonly encountered in archaeology and skyscape work in orientation and azimuth analysis (Magli, 2016; Silva, 2020). Here it becomes relevant because longitudes can be folded modulo a candidate period T : R then measures whether the folded phases concentrate toward one direction.

This is a useful diagnostic, but only a partial answer to the repeated-line question. A proximity-to-harmonics claim asks whether site coordinates fall near longitudes $\lambda_0, \lambda_0 + T, \lambda_0 + 2T, \dots$. Under modulo- T phase mapping these harmonics collapse to a common phase, so an ideal grid-aligned corpus would produce strong Rayleigh concentration. In real geographically clustered corpora, however, Rayleigh concentration and thresholded harmonic proximity can dissociate: a dominant off-harmonic cluster can produce significant Rayleigh concentration, while a corpus with many near-harmonic sites plus substantial off-harmonic mass can produce binomial enrichment without significant Rayleigh R . A non-significant Rayleigh result therefore does not rule out harmonic proximity, and a significant Rayleigh result does not by itself establish proximity to repeated longitude lines. The two statistics are complementary; they answer related but distinct questions.

The directly matched test for “sites cluster near harmonics” is straightforward. Under a uniform longitude model, the expected fraction of longitudes falling within $\pm d^\circ$ of any harmonic of period T is $p_0 = 2d/T$. This is the *geometric null*. In clustered corpora it remains a useful geometric benchmark, but the resulting p -values are interpreted here as corpus-level proximity enrichment rather than as evidence about why coordinates take those values.

This paper develops that matched test and uses three openly available monument datasets as a worked methodological example. The UNESCO dome/stupa corpus, Wikidata Q180987 stupa corpus, and OSM stupa layer are used strictly as illustration: they supply real coordinate distributions to show where Rayleigh and the geometric-null bi-

nomial agree or disagree. No substantive archaeological claim depends on the result; the worked example could in principle be replaced by any spatially clustered corpus. Under a fixed period, fixed corridor phase, fixed proximity thresholds, and fixed operational corpora, the geometric-null binomial test reports whether more sites fall near harmonic longitudes than expected from the geometric coverage of the threshold bands. Rayleigh statistics are retained as complementary circular diagnostics, and the period sweep is used to show when phase concentration and thresholded harmonic proximity agree or dissociate. The empirical application reports corpus-level proximity enrichment at the fixed 3° period and $\sim 35.25^\circ\text{E}$ Levantine corridor phase. Following [Silva \(2020\)](#), this is treated throughout as statistical description of the observed coordinate distribution in an open-data worked example; historical, cultural, or geographic explanation is a separate interpretive task requiring independent evidence.

2 Rayleigh Concentration and Harmonic Proximity: Complementary but Distinct Statistics

2.1 Why the Two Statistics Can Dissociate

Let $\lambda_1, \dots, \lambda_N$ be site longitudes and T a candidate grid period with anchor λ_0 . Define phases by $\theta_i = 2\pi((\lambda_i - \lambda_0) \bmod T)/T$, and the Rayleigh statistic by

$$R = \left| \frac{1}{N} \sum_{i=1}^N e^{i\theta_i} \right|,$$

where $i = \sqrt{-1}$ (complex exponential; equivalently, $e^{i\theta} = \cos \theta + i \sin \theta$, so R is the length of the mean of the unit vectors $(\cos \theta_i, \sin \theta_i)$).

A key property of this transformation is that sites near *any* harmonic of T all map to the same phase $\theta \approx 0$ under the modulo- T reduction. This includes sites near λ_0 , $\lambda_0 + T$, $\lambda_0 + 2T$, and so on, because $(\lambda_0 + kT - \lambda_0) \bmod T = 0$ for any integer k . Under ideal conditions, in which all sites fall exactly on a harmonic, R would equal 1. The modulo- T phase map therefore does not, by itself, produce multiple phase classes: harmonics

collapse to one class, not many.

The Rayleigh test and the binomial proximity test can nevertheless dissociate because they aggregate cross-site information differently.

What Rayleigh measures. R is the length of the mean phasor. It is sensitive to whether a *single direction* on the unit circle predominates after averaging over the full corpus. In a geographically clustered corpus, a single dominant geographic cluster can produce large R at period T simply because that cluster’s longitude maps to a concentrated phase at that particular T , regardless of whether the cluster is harmonically proximate. This is a generic property of clustered data on the unit circle; it can produce significant R at periods that have no interpretation as harmonic proximity.

What the binomial measures. The binomial proximity test counts the number of sites whose deviation from the nearest harmonic falls within a threshold d , written as $\#\{i : \delta_i \leq d\}$. With $r_i = ((\lambda_i - \lambda_0) \bmod T)$ and $\delta_i = \min(r_i, T - r_i)$, this count is independent of the mean phase direction. Sites far from harmonics (large δ_i) contribute zero to the binomial count regardless of whether they push R up or down. Conversely, a corpus can show significant binomial enrichment even when many off-harmonic sites keep the mean phasor small.

Complementarity. The two tests ask complementary questions. Rayleigh asks whether one phase direction dominates. The binomial asks whether more sites lie within a fixed reporting threshold of a harmonic than the uniform null predicts. In a real corpus these can agree or disagree, depending on the balance between harmonically proximate sites, off-harmonic sites, and the geographic clustering structure of the corpus. A non-significant Rayleigh result does not rule out harmonic proximity; a significant Rayleigh result does not establish harmonic proximity unless the phase concentration is verified to coincide with a harmonic. Relying on Rayleigh alone to evaluate a repeated-line hypothesis can therefore produce false positives at cluster-dominated periods and miss genuine proximity enrichment when near-harmonic sites coexist with substantial off-harmonic mass.

2.2 Empirical Demonstration: Period Sweep Dissociation

The following demonstration uses the UNESCO keyword-matched dome/stupa corpus described fully in Section 4. Table 1 illustrates the dissociation concretely ($N = 90$). The full sweep of 46 candidate periods (1.5° to 6.0° in 0.1° steps) is available in the supplementary audit; four representative periods are shown here to illustrate the qualitatively distinct outcomes. Both tests use the T3 threshold ($\pm 0.10 \times T$, geometric null $p_0 = 20\%$; tiers defined in Section 3.2); finer-threshold tiers (T2, T1) are reported in the primary results.

Table 1: Rayleigh concentration vs. T3 binomial proximity enrichment at four representative periods drawn from a full sweep of 46 candidate periods (1.5° to 6.0° in 0.1° steps; complete sweep tables in supplementary audit). These four periods span the full range of dissociation outcomes, including three periods with significant Rayleigh concentration but no proximity enrichment, and one period with proximity enrichment but non-significant Rayleigh concentration. UNESCO dome corpus ($N = 90$, corridor phase $\sim 35.25^\circ\text{E}$). T3 threshold = $0.10 \times T$; geometric null $p_0 = 20\%$. Sig symbols: $*p < 0.05$, $**p < 0.01$, ns = not significant.

Period	Rayleigh p	Sig	T3 hits	Expected	Binom. p	Sig
2.4°	0.0083	**	21	18.0	0.2503	ns
3.0°	0.1796	ns	29	18.0	0.0043	**
3.5°	0.0267	*	13	18.0	0.9312	ns
3.6°	0.0407	*	20	18.0	0.3383	ns

The dissociation is clear. Rayleigh is significant at 2.4° , 3.5° , and 3.6° , and non-significant at 3.0° . The binomial proximity test is significant only at 3.0° . A researcher relying on Rayleigh alone to evaluate the repeated-line hypothesis would flag periods that carry no proximity enrichment, and would overlook the period with the clearest proximity structure. Using the two tests together avoids this.

The table shows how the two diagnostics differ. At 2.4° , 3.5° , and 3.6° , one geographic cluster dominates the phase distribution; these periods show concentrated R without harmonic proximity enrichment. At 3.0° , many sites fall near the harmonic class, but substantial off-harmonic mass keeps the mean phasor small; the binomial test reports the thresholded per-site proximity counts. Rayleigh and the binomial measure genuinely different aspects of the data; neither is wrong, but only the binomial directly answers the

question “do sites fall near harmonics?”

A broader candidate-period survey of the dome/stupa corpus (1.5° to 6.0° in 0.1° steps, 46 periods; four representative periods shown in Table 1) reports the dominant binomial proximity peak at 3° ($p = 0.0043$); selected nearby non-harmonic spacings (1.8°, 2.1°, 2.7°, 3.3°) are non-significant in the same sweep. This sets 3° as the primary period for the cross-corpus analysis in Section 5.

The dissociation also appears in the full UNESCO cultural corpus ($N = 1,011$ sites with coordinates, including non-dome cultural sites). At $T = 3^\circ$, Rayleigh remains non-significant ($p = 0.6008$), while binomial proximity enrichment is significant across all three tier thresholds (T3: $p = 0.0246$; T2: $p = 0.0011$; T1: $p = 0.0103$). Because this corpus includes the dome sites that set the period, it is not a cross-corpus check. It shows that the dissociation between Rayleigh and binomial results is also observed outside the restricted dome vocabulary.

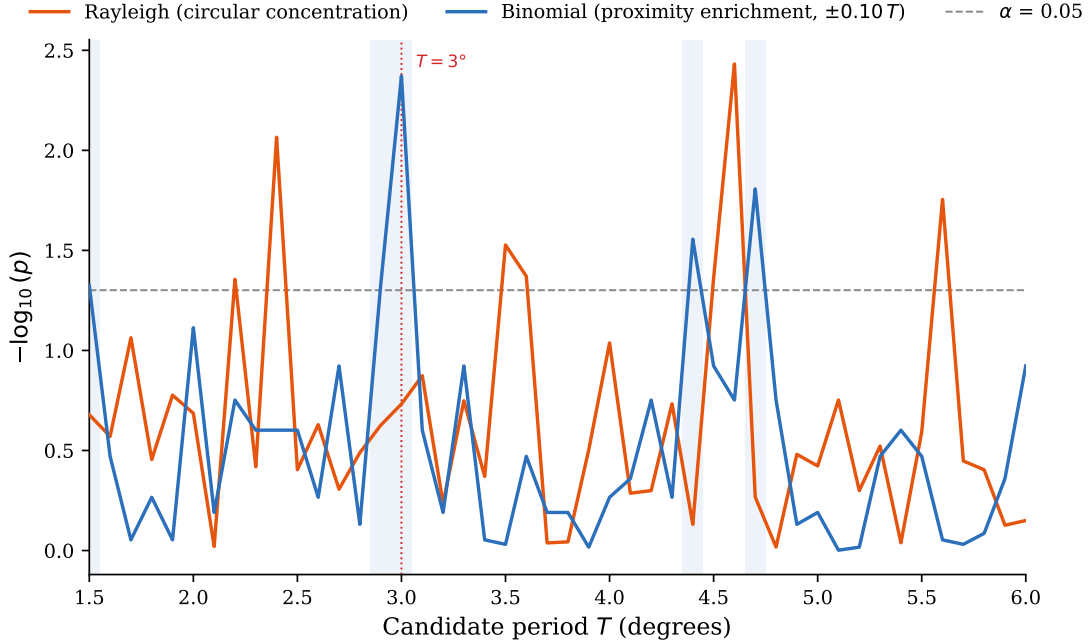


Figure 1: Period sweep of the UNESCO dome/stupa corpus ($N = 90$): $-\log_{10}(p)$ for the binomial proximity-enrichment test (blue; threshold $\pm 0.10 \times T$) and the permutation Rayleigh test (orange) across 46 candidate periods (1.5° to 6.0° in 0.1° steps); anchor $\approx 35.25^\circ\text{E}$. Dashed line marks $\alpha = 0.05$. The blue shaded column at $T = 3^\circ$ highlights the dominant proximity peak; Rayleigh is non-significant at this period, illustrating the dissociation between Rayleigh and binomial results.

A Monte Carlo spacing test (100,000 permutations, dome label shuffle) across 18

candidate spacings from 0.5° to 15° (full list in supplementary audit) finds the observed count of 4 significant spacings marginal ($p \approx 0.0979$). Three of the four significant spacings (6° , 8° , 12°) are multiples of 3° and their hits are fully accounted for by the 3° harmonic structure; the 3° spacing is the only one whose significance is not accounted for by overlap with another spacing’s nodes.

3 The Geometric-Null Binomial Framework

3.1 The Geometric Null

For a grid of period T anchored at λ_0 , define the anchor-relative remainder and nearest-harmonic deviation as

$$r_i = ((\lambda_i - \lambda_0) \bmod T), \quad \delta_i = \min(r_i, T - r_i).$$

A site falls within tier threshold d if $\delta_i \leq d$. Under a uniform longitude model, the expected fraction of longitudes satisfying that inequality is $p_0 = 2d/T$. In clustered corpora this remains a useful geometric benchmark; the resulting p -values are interpreted as corpus-level proximity relative to the stated geometric benchmark, not as evidence about why sites are located where they are. Declustering is not applied because the target of description is the observed operational corpus, not an analyst-thinned version of it. Geographic clustering is a feature of the coordinate distribution being described; it is neither interpreted nor removed. Removing or down-weighting clustered sites would answer a different question about a transformed dataset. The multi-corpus comparison is descriptive: the three corpora have different geographic nodes (the distributed Eurasian corridor for UNESCO, the Java/Sumatra node for Wikidata, and crowd-sourced coverage across South and Southeast Asia for OSM), and the same frame-conditional proximity pattern is observed across the differently assembled datasets. As a sensitivity check, the Wikidata binomial test was rerun with all sites within a 2° bounding box centred on Borobudur ($\sim 110^\circ\text{E}$) excluded; proximity enrichment persists at T2 and T3 (details in

Zenodo audit). Two control corpora provide descriptive boundary cases: the UNESCO Americas sub-corpus shows no proximity enrichment at the Levantine phase (T2 rate below the geometric null), and the North American mound corpus shows no enrichment at any tier (Section 5).

The two null models reported in Section 5 (Null A: within-dome bootstrap; Null B: same-footprint UNESCO background, the discriminating comparison) provide complementary assessments of the observed UNESCO proximity pattern; they are discussed in Section 6.

3.2 Tier Classification

We define three proximity thresholds as round fractions of the $T/2$ half-period (1.5°), calibrated to bracket the empirical natural proximity scales of the corpora under study: the Old World Trade Routes (OWTRAD; Ciolek 2004) Silk Road network (2.6%), the UNESCO dome subset (3.6%), the full UNESCO Cultural/Mixed corpus (10.2%), and the Wikidata stupa corpus (20.2%) (peak-enrichment null rates from the multiscale window sweep; reproduction in `supplementary/audit/tier_calibration_audit.py`). Each threshold corresponds to a geometric null rate $p_0 = 2d/T$ (T1 denotes the finest, strictest threshold; T3 the most lenient):

Tier	Threshold	Geometric null p_0
T1	$\delta \leq 0.06^\circ$ ($\lesssim 6.66$ km equatorial)	4%
T2	$\delta \leq 0.15^\circ$ ($\lesssim 16.6$ km)	10%
T3	$\delta \leq 0.3^\circ$ ($\lesssim 33$ km)	20%

Tier labels are descriptive shorthand; the primary test is always the binomial. Because the thresholds are calibrated to the empirical proximity scales of the corpora under study, data-adaptive threshold selection of this kind may favor detecting enrichment at the chosen values; results should be evaluated by the binomial statistics rather than the tier labels themselves.

3.3 Cross-Corpus Consistency via Fisher Combination

For each corpus c at tier d , a one-sided binomial test yields p_c . Fisher’s combination statistic $\chi^2 = -2 \sum_c \ln p_c$, referred to $\chi^2(2C)$ for C corpora, summarizes whether the directional excess is consistent across all corpora simultaneously. Fisher’s method formally assumes independent p -values. The three corpora were assembled by distinct processes and their geographic clusters are substantially non-overlapping, but they share historical and geographic structure (Eurasian corridor, stupa tradition, public data sources). The combined p -value is therefore used as a descriptive consistency summary only; the individual corpus p -values are the primary quantities reported, and the combination summarizes their direction rather than providing a formal joint test with frequentist error guarantees.

The dome corpus period sweep reports 3° as the dominant proximity peak in that corpus; the differently assembled corpora are evaluated at the same period. Wikidata and OSM were evaluated at 3° without any period sweep of those corpora. This asymmetry is verifiable from the published code and audit files. Following [Silva \(2020\)](#), all results are treated as statistical description of the observed coordinate distribution; there is no prior documentation of a 3° harmonic grid anchored to the Levantine corridor, and these statistics do not identify such a grid absent independent archaeological support. The two-corpus Fisher restricted to Wikidata and OSM (reported in Table 2 as “Dome-excl. combined”) summarizes whether the same frame-conditional proximity pattern is observed across differently assembled datasets.

4 Data

UNESCO dome/stupa subset ($N = 90$). The UNESCO World Heritage XML dataset ([UNESCO World Heritage Centre, 2024](#)) contains 1011 Cultural and Mixed inscribed sites. Seven morphological keywords (stupa/stupas, tholos, dome/domed/domes, spherical) were matched against site names, XML descriptions, and OUV text, yielding $N = 90$ sites. All statistical tests use this keyword-matched population.

Wikidata Q180987 stupa corpus ($N = 229$). All *instance-of*: *stupa* (Q180987) items in the Wikidata knowledge base, extracted via SPARQL query on the *instance-of*: *stupa* property. The corpus is geographically concentrated in the Java/Sumatra node, roughly from $\sim 105^\circ\text{E}$ to $\sim 115^\circ\text{E}$ (Miksic, 1990).

OSM stupa layer ($N = 259$). OpenStreetMap features with tags `building=stupa`, `historic=stupa`, or `ruins=stupa`, assembled by crowd-sourced tagging via Overpass API (OpenStreetMap contributors, 2024). Data © OpenStreetMap contributors, available under the Open Database Licence (ODbL 1.0; <https://opendatacommons.org/licenses/odbl/>). The corpus is treated as a co-primary dataset: it is assembled by a process entirely independent of UNESCO and Wikidata curation, and OSM’s coverage notes are a property of the dataset as a whole rather than a limitation specific to this application.

All three corpora were tested against the same fixed candidate frame: period 3° and the $\sim 35.25^\circ\text{E}$ Levantine corridor phase. Because open datasets are publicly available, the frame is treated uniformly as fixed for the present analysis across all three corpora. The empirical application therefore functions as a frame-specified spatial test.

Candidate frame. The candidate reference phase is the $\sim 35.25^\circ\text{E}$ Levantine corridor phase, which emerged as the dominant proximity peak phase in the dome corpus sweep described in Section 2; Wikidata and OSM were evaluated at this phase without independent sweeps of their own. The counts are invariant within the band. Any anchor longitude within $\pm 0.06^\circ$ of this phase, including Jerusalem at 35.232°E , produces identical proximity counts, and the data cannot distinguish between longitudes on the same grid within the same proximity tier. The anchor is labeled by Mount Gerizim at 35.269°E , a topographic summit within the Levantine corridor absent from all monument corpora and not derived from any external literature on longitude grids. It serves as a named geographic coordinate for the interval under test; no substantive weight should be attached to the choice of Gerizim over any other co-grid longitude within the same proximity tier.

5 Results

5.1 Primary Result: Cross-Corpus Geometric-Null Test

Table 2 reports the primary statistical results: one-sided binomial tests and Fisher combinations across all three corpora.

Table 2: Cross-corpus geometric-null harmonic-proximity enrichment at the fixed 3° period and $\sim 35.25^\circ\text{E}$ Levantine corridor phase. For each corpus and tier, the one-sided binomial test compares the observed number of sites within distance d of any harmonic with the expected count under the geometric benchmark $p_0 = 2d/T$. The “Dome-excl. combined” row restricts Fisher’s combination to Wikidata and OSM (excluding the period-motivating dome corpus) and is the primary cross-corpus consistency summary; see Section 3 for the independence caveat.

Tier	Corpus	N	Hits	Exp	Fold	p_{binom}	Sig
T1 ($p_0=4\%$)	UNESCO domes	90	11	3.6	$3.06\times$	0.0009	***
	Wikidata stupas	229	11	9.2	$1.2\times$	0.311	ns
	OSM stupas	259	12	10.4	$1.16\times$	0.3433	ns
	<i>All-3 Fisher $\chi^2(6) = 18.402$, nominal $p = 0.005$ (includes period-motivating corpus; for completeness)</i>						
	<i>Dome-excl. combined (Wiki+OSM) $\chi^2(4) = 4.474$, $p = 0.346$, ns</i>						
T2 ($p_0=10\%$)	UNESCO domes	90	18	9.0	$2.0\times$	0.0032	**
	Wikidata stupas	229	33	22.9	$1.44\times$	0.0212	*
	OSM stupas	259	33	25.9	$1.27\times$	0.089	
	<i>All-3 Fisher $\chi^2(6) = 24.017$, nominal $p < 0.001$ (includes period-motivating corpus; for completeness)</i>						
	<i>Dome-excl. combined (Wiki+OSM) $\chi^2(4) = 12.547$, $p = 0.014$, *</i>						
T3 ($p_0=20\%$)	UNESCO domes	90	29	18.0	$1.61\times$	0.0043	**
	Wikidata stupas	229	70	45.8	$1.53\times$	0.0001	***
	OSM stupas	259	69	51.8	$1.33\times$	0.0059	**
	<i>All-3 Fisher $\chi^2(6) = 39.661$, nominal $p < 0.001$ (includes period-motivating corpus; for completeness)</i>						
	<i>Dome-excl. combined (Wiki+OSM) $\chi^2(4) = 28.74$, $p < 0.001$, ***</i>						

In the worked example, all three corpora exceed the geometric null at every tier despite substantially different geographic footprints and assembly processes. Fisher combinations are reported as cross-corpus consistency summaries (see Section 3 for the independence caveat). The UNESCO T1 result shows the largest individual fold-enrichment ($3.06\times$); Wikidata and OSM T1 results are individually non-significant. The two-corpus combination restricted to Wikidata and OSM (“Dome-excl. combined”) yields $\chi^2(4) = 28.74$, $p < 0.001$ at T3 and $\chi^2(4) = 12.547$, $p = 0.014$ at T2; the T1 two-corpus combination is non-significant ($p = 0.346$). (The all-three-corpus rows in the table include the dome corpus and are reported for completeness only; because the dome corpus motivated the

period they are not informative as cross-corpus consistency checks.) All p -values are frame-conditional proximity-enrichment measures for the fixed 3° period and $\sim 35.25^\circ\text{E}$ Levantine corridor phase; following [Silva \(2020\)](#) they are statistical descriptions of the observed coordinate distribution, not evidence of historical placement. The 3° period emerged as the dominant proximity peak in the dome corpus sweep (Section 2); the three-corpus analysis is the methodological illustration that the same frame-conditional proximity pattern is observed across differently assembled datasets at this period.

5.2 Supplementary Rayleigh Phase Concentration

Rayleigh statistics are retained as complementary diagnostics for comparability with prior literature. Table 3 reports them at $T = 3^\circ$. The full UNESCO corpus shows no Rayleigh phase concentration ($R = 0.0224$, perm- $p = 0.6008$, ns) at $T = 3^\circ$, consistent with the proximity pattern not being a generic circular-statistical property of the full heritage list. Watson’s U^2 and Kuiper tests are retained as supplementary two-sample diagnostics (not independent tests): both show that the dome/stupa phase distribution differs detectably from the full UNESCO corpus ($p = 0.0036$, $p = 0.0013$ respectively). Because any geographically concentrated subset will tend to differ from its parent corpus in phase distribution, these results show concentration rather than proximity enrichment per se; they are included for comparability with prior literature that uses two-sample circular diagnostics as subset-specificity checks.

Table 3: Supplementary Rayleigh phase concentration at $T = 3^\circ$, retained for comparability with prior literature. The binomial proximity result (Table 2) is the matched test; Rayleigh statistics are reported here as supplementary diagnostics. All results are conditional on the fixed corridor-phase frame.

Corpus	N	R	perm- p	Sig
UNESCO full corpus	1011	0.0224	0.6008	ns
UNESCO dome/stupa	90	0.1365	0.1883	ns
Wikidata Q180987	229	0.1669	0.0016	**
OSM stupa layer	259	0.1073	0.0355	*
<i>Fisher (Wikidata + OSM): $\chi^2(4) = 19.551$, $p = 0.0006$</i>				

5.3 Route-Network Contrast, Negative Controls, and Scope

OWTRAD Silk Road route-network contrast. The OWTRAD Silk Road route network (1674 nodes, 1946 route segments) is included as an exploratory contrast. It is not a negative control: the Silk Road network occupies the same geographic range as the monument corpora, and proximity enrichment cannot be assumed absent without testing. Two binomial proximity tests are distinguished by their sampling unit. The per-segment midpoint test uses one longitude per route segment (non-deduplicated), weighting each segment equally regardless of how many edges meet at a hub; this test is non-significant at both tiers (T1: $p = 0.0908$, ns; T2: $p = 0.0679$, ns). The degree-weighted node test weights each node by its degree centrality (the number of incident route edges), as a proxy for hub importance; this test is significant at both tiers (T1: $p \leq 0.0001$; T2: $p = 0.0002$). The degree-weighted result is itself an additional positive proximity count: Eurasia spans roughly 180 of longitude, and the result concerns proximity to a specific 3 harmonic within that space. This result therefore does not function as a negative control and is not explained by the coordinate data; it is discussed further in Section 6. The NPC directional test does not show the monument phase ($K = 0.7429$, perm- $p = 0.46532$, ns), indicating that the route-segment phase distribution as a whole does not share the monument mean phase direction.

The UNESCO Americas sub-corpus ($N = 145$) shows T2 rate of 9.7%, below the 10% geometric null, indicating no proximity enrichment at the $\sim 35.25^\circ\text{E}$ Levantine corridor phase in the Western Hemisphere.

A type-matched geographic control uses the North American mound corpus ($N = 236$): National Historic Landmark (NHL) and National Register of Historic Places (NRHP)-registered earthen burial and platform mounds assembled from the NPS National Register database with coordinates resolved via Wikidata (NPS, 2024; Wikimedia Foundation, 2024). These are broadly mounded earthen structures used here as a geographically distinct type control, with coordinates entirely in the Western Hemisphere outside any connection to the $\sim 35.25^\circ\text{E}$ Levantine corridor phase. At the 3° period and $\sim 35.25^\circ\text{E}$ Levantine corridor phase the mounds corpus shows no proximity

enrichment at any tier: T1: 1/236 (0.4%, null 4%, $p = 1.00$, ns); T2: 11/236 (4.7%, null 10%, $p = 1.00$, ns); T3: 45/236 (19.1%, null 20%, $p = 0.66$, ns). All three tiers are at or below the geometric null. This provides a descriptive boundary case: the reported proximity pattern is observed in the Eurasian corpora here and not in this Western Hemisphere type control.

Null B (same-footprint restricted draw, 100,000 trials) is the discriminating comparison: does the dome/stupa subset show greater harmonic proximity than the broader UNESCO background occupying the same longitude footprint? This comparison is possible for the UNESCO corpus because a larger parent population (the full UNESCO Cultural/Mixed list, $N = 1011$ sites) exists from which the dome/stupa subset was selected. No equivalent parent population exists for the Wikidata or OSM corpora, which are already operationally defined as all stupa-tagged items; a same-footprint Null B comparison therefore applies only to the UNESCO corpus. Three window widths were tested ($\pm 2^\circ$, $\pm 5^\circ$, $\pm 10^\circ$; Bonferroni $\alpha_{\text{adj}} = 0.0167$): the $\pm 2^\circ$ window gives $p = 0.0180$; the $\pm 5^\circ$ window gives $p = 0.0129$; the $\pm 10^\circ$ window gives $p = 0.0139$. All three windows survive Bonferroni correction ($\alpha_{\text{adj}} = 0.0167$), illustrating subset-specific enrichment within the shared geography across all window choices. A within-dome bootstrap (Null A, 100,000 trials) resamples from the dome sites' own longitude distribution; its expected hit rate is defined by the same observed coordinate distribution, so it cannot serve as a discriminating comparison. The observed T1 count is non-significant under this null ($p = 0.5493$, ns), consistent with the proximity pattern being conditional on the dome-site longitude distribution.

5.4 Anchor Concordance

All three corpora show the same $\sim 35.25^\circ\text{E}$ Levantine corridor phase. The Normalized Phasor Consensus statistic K is defined as $K = |\bar{u}_A \cdot \bar{u}_B|$, where $\bar{u}_c = (\sum_k e^{i\theta_k^{(c)}}) / |\sum_k e^{i\theta_k^{(c)}}|$ (unit complex numbers as in Section 2) is the normalized mean phasor of corpus c at the candidate period, and $|\cdot|$ denotes the modulus of the dot product of two unit complex numbers. The dot-product formulation is preferred over the circular correlation coefficient

(Mardia & Jupp, 2000, §8.3) because it compares phase *directions* only, stripping out amplitude information that is irrelevant when corpora differ substantially in size and clustering: $K = 1$ when the two corpora point to the same phase, $K = 0$ when they are orthogonal. K is introduced here as an original statistic for this application; there is no prior citation. Its null distribution is estimated by permutation: site longitudes within each corpus are independently and uniformly reshuffled (each corpus permuted separately, with no cross-corpus label exchange), the phasor direction $\bar{u}_c$ is recomputed for each permuted corpus, and K is recalculated; this is repeated 100,000 times to obtain the reference distribution of K under independent random longitude assignments. The resulting p -value is the fraction of permuted draws with K at least as large as the observed value. Full permutation code is in the Zenodo audit. In the worked example, $K \geq 0.9675$ for all pairings of the three monument corpora (Table 4). The OSM + Wikidata pair ($K = 0.9695$, perm- $p = 0.15884$, ns) illustrates the power limitation of two-corpus tests when both corpora share a Southeast Asian geographic footprint: the high K indicates strong phase agreement, but the permutation null cannot be rejected with two corpora alone. The three-corpus combination reaches perm- $p = 0.02437$ with consensus anchor at 0.029° from 35°E . The primary non-nested two-corpus pair (Dome + Wikidata) is individually non-significant, reflecting the limited power of a two-corpus permutation test: high K indicates strong phase agreement, but two corpora alone cannot reject the null that such agreement arises by chance. With the OSM corpus added as a third data source, the three-corpus statistic reaches significance: the same phase directions are uncommon under the permutation null even when individual pairs are underpowered. Negative controls show K drops sharply when the monument corpora are replaced with geographic or route-network controls.

Table 4: Selected NPC anchor-concordance configurations at $T = 3^\circ$ (100,000 permutation draws). The “Conservative” row pairs the broadest possible corpus (full UNESCO Cultural/Mixed) with the Wikidata stupa corpus, maximizing denominator size at the cost of corpus specificity. Negative controls (the final three rows) indicate K is not a generic Eurasian property.

Role	Corpora	K	perm- p	Note
Non-nested pair	Dome + Wikidata Stupa	0.9675	0.16151	Primary non-nested
Non-nested pair	OSM + Wikidata Stupa	0.9695	0.15884	SE-Asia geography; underpowered
Conservative	UNESCO full + Stupa	0.982	0.11864	Broadest corpus pair
Three-corpus	Dome + Stupa + OSM	0.972	0.02437	All 3 monument corpora
Neg. control	Americas + Wikidata Stupa	0.7028	0.50455	Geographic control
Contrast corpus	UNESCO + OWTRAD	0.7429	0.46532	Route network
Type control	N. American mounds ($N=236$)	n/a	n/a	No proximity enrichment

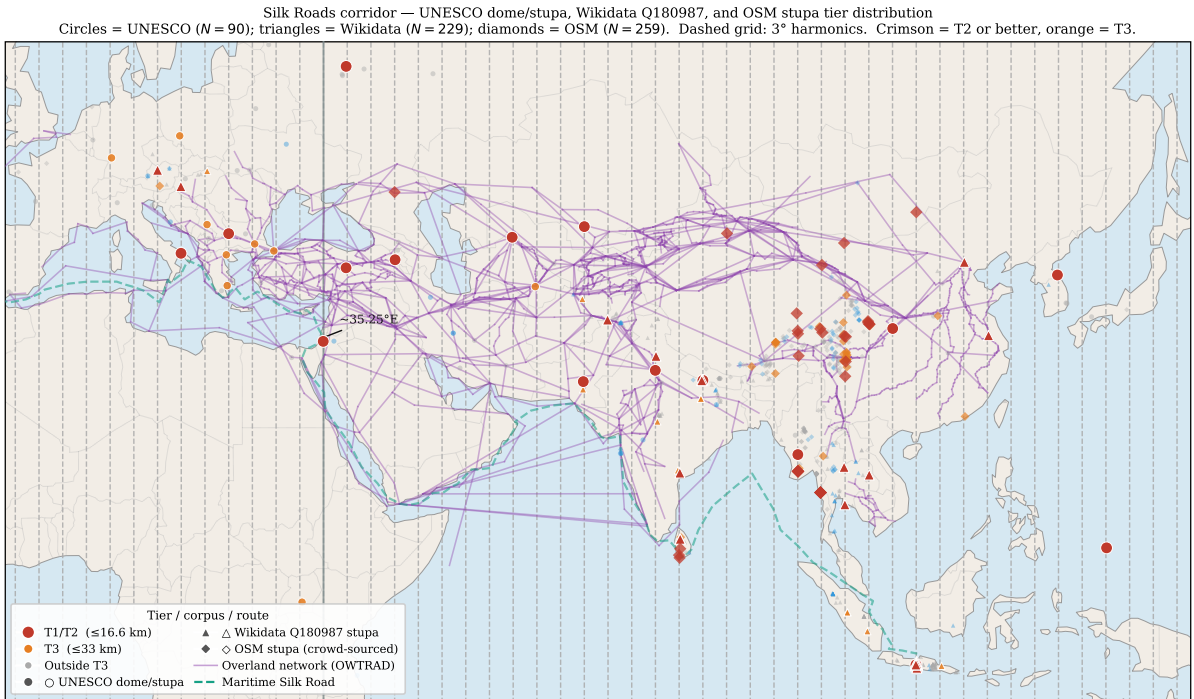


Figure 2: Silk Roads corridor showing all three monument corpora mapped against the 3° harmonic grid (dashed lines) anchored to the $\sim 35.25^\circ\text{E}$ Levantine corridor phase. Circles = UNESCO dome/stupa ($N = 90$, keyword-matched); triangles = Wikidata Q180987 stupa ($N = 229$); diamonds = OSM stupa ($N = 259$, crowd-sourced). Color indicates harmonic proximity tier: crimson = T2 or better (≤ 16.6 km), orange = T3 (≤ 33 km), gray = outside T3. The dark meridian marks the fixed $\sim 35.25^\circ\text{E}$ corridor phase. Teal dashed lines show the maritime Silk Road route layer, digitised from harbour attestations in Hansen (2012), Sen (2014), and Tomber (2008). All results are conditional on the fixed corridor-phase frame. Map projection: Mercator (WGS 84). Base map: Natural Earth (public domain, <https://www.naturalearthdata.com/>, accessed 1 March 2025).

6 Discussion

6.1 The Methods Contribution

The central methodological result is the separation between circular phase concentration and thresholded harmonic proximity. Rayleigh R describes whether the full phase distribution has a dominant mean direction at period T ; the geometric-null binomial describes whether a corpus contains more sites within a stated operational distance of any harmonic than expected from the fraction of longitude covered by those bands. Because the two statistics weight off-harmonic observations differently, they provide complementary information and can dissociate within the same corpus. This makes Rayleigh a useful diagnostic for phase structure and the geometric-null binomial the matched statistic for thresholded proximity-to-harmonics claims. In Silva’s terms, the tests detect and describe structured features in the spatial data; archaeological, historical, or geographic explanations for those features remain interpretive questions to be evaluated with independent evidence.

The analysis is a statistical description of harmonic proximity in the observed coordinate distribution. Geographic clustering is part of the documented corpus pattern, so site-level binomial p -values describe corpus-level proximity enrichment rather than site-location decisions. The geometric null $p_0 = 2d/T$ gives a transparent expected hit fraction under uniform longitude, and Fisher’s method is used as a cross-corpus consistency summary. This framework is applicable to any spatial periodicity hypothesis of this form regardless of geographic or cultural context.

6.2 Interpretation of the Empirical Illustration

The three-corpus worked example reports directional excess over the geometric null at the fixed 3° period and $\sim 35.25^\circ\text{E}$ Levantine corridor phase. The dome-excluded two-corpus combination (Wikidata + OSM) yields a Fisher combination of $\chi^2(4) = 28.74$, $p < 0.001$ at T3, and is the primary cross-corpus result reported here. The all-three-corpus combination (Fisher $p < 0.001$ at T3) is reported in Table 2 for completeness only; it

includes the UNESCO dome corpus, which partly motivated selection of this period, and is therefore a less conservative benchmark. All results are frame-conditional proximity-enrichment measures (see Limitations). The UNESCO corpus is the most geographically distributed result, with T3 sites spanning the full Eurasian corridor.

Null B (same-footprint restricted draw) provides the discriminating comparison: among all UNESCO cultural sites occupying the same longitude footprint, the dome/stupa subset shows greater harmonic proximity than the broader background across all three window widths after Bonferroni correction (see Results), illustrating subset-specific enrichment within the shared geography. The cross-corpus phase concordance (NPC $K \geq 0.9675$ across all pairings of three independently assembled monument datasets with different geographic centers of mass) is a frame-conditional diagnostic: UNESCO dome sites, Wikidata Q180987 stupas, and OSM stupa tags show the same $\sim 35.25^\circ\text{E}$ phase pattern in the reported coordinate data.

Corpus overlap and curation. The three corpora are assembled by distinct processes: the UNESCO dome/stupa subset derives from keyword matching against the WHC XML dataset; Wikidata Q180987 was extracted via SPARQL query on the *instance-of: stupa* property; OSM was assembled by crowd-sourced tagging of `building=stupa`, `historic=stupa`, and `ruins=stupa` features via Overpass API. An empirical overlap check using GPS proximity (threshold < 1 km) and name-string matching finds 4 near-coincident site pairs between UNESCO and Wikidata, 0 between UNESCO and OSM, and 1 between Wikidata and OSM (see supplementary overlap audit). Because all three datasets are publicly available, the candidate frame is handled as fixed across the empirical demonstration. The Fisher combination is used as a consistency check across differently assembled datasets. The geographic clusters of the three corpora are substantially non-overlapping at the region level: UNESCO dome sites span the full Eurasian corridor from the Levant to Java; Wikidata is dominated by the Java/Sumatra node; OSM coverage reflects crowd-sourced mapping density across South and Southeast Asia. The same corpus-level proximity pattern is therefore not confined to a single shared geographic cluster.

The Wikidata result is dominated by the Java/Borobudur geographic cluster (Miksic, 1990): that cluster’s position on a 3° harmonic is part of the proximity structure the test reports. The relevant cross-corpus check is whether the same pattern appears in differently assembled datasets whose footprints do not overlap with Borobudur. The five near-coincident pairs identified across all three corpora (4 UNESCO–Wikidata, 0 UNESCO–OSM, 1 Wikidata–OSM) are all harmonic-proximate sites; removing them does not change any binomial hit count and therefore does not affect any reported p -value. The OSM result, assembled by a process with no relationship to UNESCO or Wikidata curation, shows the same frame-conditional proximity pattern ($p = 0.0059$ at T3) and the same phase concordance with Wikidata ($K = 0.9695$) in the worked example. Figure 2 shows all three corpora mapped against the 3° harmonic grid.

The interpretation follows Silva (2020)’s distinction between statistical description and archaeological explanation. Significance tests report departures from a null model; archaeological explanation requires evidence independent of the coordinates themselves, including the wider archaeological record and socio-historical context. The result should be read literally. In the reported open-data corpora, more sites fall near the specified 3° harmonic frame than the geometric benchmark predicts. The analysis does not explain why the sites are located where they are and does not identify a cultural, religious, astronomical, trade, or transmission mechanism. Those questions require evidence outside the coordinate data analyzed here. An additional positive finding is reported descriptively: weighting OWTRAD nodes by degree centrality (the number of route edges incident to each node, as a proxy for hub importance) finds that the most network-central Silk Road cities also fall near the 3° harmonic grid more often than the uniform null predicts. Eurasia spans roughly 180° of longitude, and the finding concerns proximity to a *specific* 3° harmonic within that space. Whether the monument result and the OWTRAD result are statistically independent cannot be determined from these data. The degree-weighted OWTRAD finding is reported here as an open descriptive observation, not as support for any interpretive claim; it is included because omitting a positive finding in an openly reproducible analysis would be selective reporting. Historical interpretations involving

deliberate placement would additionally require evidence for pre-modern continental longitude determination at sub-degree precision (Hunger & Steele, 2019; Neugebauer, 1975). The contribution of the present work is to establish what the directly matched statistical test reports when applied to three differently assembled corpora.

6.3 Limitations

Frame conditionality. All p -values in Table 2 are conditional on the fixed 3° period and $\sim 35.25^\circ\text{E}$ Levantine corridor phase (anchored to Mount Gerizim, 35.269°E). All three corpora are evaluated against this same frame. The results are therefore corpus-level proximity-enrichment measures under that fixed frame and should not be read as outcomes of an unconditional search over all possible periods and anchors.

Analysis sequence and exploratory scope. This paper is an exploratory methods contribution following Silva (2020). Pre-registration is a standard for confirmatory hypothesis tests; no registry exists for, and none is expected of, open-data descriptive spatial analyses of this kind. The appropriate sequence for exploratory work is to identify a statistical pattern in one dataset and assess whether it is observed across differently assembled datasets, which is what the dome-corpus sweep followed by the Wikidata+OSM cross-check does. The Wikidata+OSM Fisher combination is the primary cross-corpus summary precisely because it excludes the period-motivating corpus; the three-corpus result is a less conservative supplement. The asymmetry is fully disclosed and verifiable from the published code.

Anchor-sweep correction and its implications. A max-statistic sweep of 36,000 trial longitudes is run against the full UNESCO Cultural/Mixed corpus ($N = 1011$), not the dome subset, using the T2 proximity threshold ($\delta \leq 0.15^\circ$, 10% geometric null). The mean T2 hit count across all 36,000 trial anchors is 101.2 ± 10.9 , close to the expected 10% of $N = 1011$. The full heritage list is geographically clustered worldwide and no phase stands out strongly when measured against the entire list. Gerizim ranks at the 99.7th percentile ($p = 0.0753$), placing it in the top 2.3% globally. The anchor-sweep correction asks whether the Levantine phase is an unconditional max-statistic outlier

over the entire heritage list. On that diagnostic, the phase is not reported here as an unconditional discovery from the UNESCO corpus after anchor-search correction. The phase was identified from the dome corpus sweep and then held fixed; we do not claim it is uniquely exceptional among all anchors in the full heritage list. The Wikidata and OSM corpora were evaluated at this pre-specified frame, so their binomial results are frame-conditional diagnostics rather than unconditional anchor-search results, and the primary cross-corpus result is the Wikidata+OSM Fisher combination, which excludes the corpus from which the frame was derived. Full anchor-sweep details are in the Zenodo audit.

7 Conclusion

This paper shows the distinction between Rayleigh concentration and thresholded harmonic proximity as descriptors of a modulo- T phase distribution: Rayleigh summarizes mean circular concentration, while the geometric-null binomial tests thresholded proximity to any harmonic under the benchmark $p_0 = 2d/T$. The period sweep illustrates this distinction empirically, with Rayleigh-significant periods lacking proximity enrichment and the enriched 3.0° period lacking significant Rayleigh concentration. Applied under a fixed operational frame, with fixed period, corridor phase, thresholds, and corpora, the binomial framework reports a consistent corpus-level proximity pattern across three hemispherical-monument datasets at the 3° period and $\sim 35.25^\circ\text{E}$ Levantine corridor phase.

Two qualifications contextualize these results, both detailed in the Limitations section. The period emerged from the dome corpus sweep; the Wikidata+OSM combination ($\chi^2(4) = 28.74$, $p < 0.001$ at T3) is the primary cross-corpus check precisely because it excludes that corpus.

Following [Silva \(2020\)](#), the worked example is best read as a descriptive spatial exercise. This paper is a methods contribution: it introduces and demonstrates the geometric-null binomial framework, shows that Rayleigh concentration and geometric-null binomial

proximity can dissociate within the same corpus while each statistic remains useful for its own question, and uses three hemispherical-monument corpora as a worked example to illustrate how the method behaves on real data. No confirmatory historical judgement is made, and none is possible from these results alone. The statistics report a corpus-level proximity pattern under a fixed frame; they cannot identify causal or intentional relationships. This is an open-data observational analysis: the datasets are publicly available and every step is independently reproducible from the published code and audit files. The evidentiary gap between “sites are disproportionately close to a 3° harmonic grid” and “sites were placed with knowledge of that grid” lies outside the scope of any purely statistical test; such claims require independent archaeological and socio-historical evidence that proximity statistics cannot supply. It also provides a reproducible open-data example that future comparative work can evaluate.

All data, code, and audit files are openly available at Zenodo (Tenenbaum 2026; <https://doi.org/10.5281/zenodo.20123709>).

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Data Availability

All data, code, and audit files: Zenodo (Tenenbaum 2026; <https://doi.org/10.5281/zenodo.20123709>, confirmed resolving May 2026). UNESCO XML (UNESCO World Heritage Centre, 2024).

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CRedit Author Statement

Seth Tenenbaum: Conceptualization, Methodology, Software, Formal analysis, Data curation, Writing: original draft, Writing: review & editing, Visualization.

586 Declaration of Generative AI and AI-Assisted Technolo- 587 gies in the Writing Process

588 During the preparation of this work the author used Claude (Anthropic) to assist with
589 prose drafting and GitHub Copilot (Microsoft) to assist with code autocompletion. After
590 using these tools, the author reviewed and edited all content as necessary and takes full
591 responsibility for the content of the published article.